



Scientific Notation

Day 5 of 20 · Duration: 50 minutes · Theme: Itasca State Park

MINNESOTA STANDARDS (MDE 2022)

Standard: 9.3.5.1

Add, subtract, multiply and divide numbers in scientific notation. (MP6, MP7)

ESSENTIAL QUESTIONS

- How do we write very large and very small numbers more compactly?
- What does the exponent in 10^n tell us about the size of a number?
- How is scientific notation useful when working with real-world quantities?

VOCABULARY

Scientific Notation — A way to write very large or very small numbers as a coefficient times a power of ten. **Coefficient (Sci. Not.)** — The number in front of the power of ten, written between 1 and 10. **Power of Ten** — 10 raised to an exponent; 10^6 has 6 zeros when written out.

LEARNING OBJECTIVES & SUCCESS CRITERIA

- I can represent very large and very small numbers in scientific notation as a coefficient times a power of ten.
- I can show my work step by step and check my answers for reasonableness.
- I can explain my reasoning to a partner using math vocabulary.

ENGAGEMENT STRATEGY

Strategy: Think-Write-Pair-Share

Practice Format: Merit Time Game — Power of 10 Lane

Expectation: Every student responds on a whiteboard or to a partner before any answer is shared.

BELL RINGER — WHITEBOARDS READY!

Prompt: Write 4,000,000 using a power of ten, then write 3.2×10^4 in standard form. *Students write their response on a whiteboard and hold it up. Teacher scans for misconceptions before moving on.*

LESSON PACING (50 MINUTES)

Phase 1: Bell Ringer 5 min min

Write 4,000,000 using a power of ten, then write 3.2×10^4 in standard form.

Phase 2: Direct Instruction 10 min min

Scientific Notation — teacher-led notes (I Do); GEMDAS reference as needed.

Phase 3: Guided Practice 12 min min

Work together (We Do) — Merit Time Game — Power of 10 Lane. Solve together: Write 3.2×10^4 in standard form. -> Answer: 32,000.

Phase 4: Culture Connection 8 min min

Ethiopian (Ethiopia) — Ethiopian Americans are a growing Minnesota community, centered in Cedar-Riverside. Ethiop (with math word problem)

Phase 5: Practice & Game 10 min min

GO: Write 3.2×10^4 in standard form. | SET: Write 0.00052 using scientific notation. Multiply 2×10^3 by 3×10^4 . | EXTRA: A cell phone has 6.4×10^{10} bytes of storage. Write it in standard form.

Phase 6: Exit Ticket 5 min min

Write 0.00052 using scientific notation, then write 2×10^7 in standard form.

PRACTICE PROBLEMS (GO / SET / EXTRA)

GO: Write 3.2×10^4 in standard form. **Answer:** 32,000

SET: Write 0.00052 using scientific notation. Multiply 2×10^3 by 3×10^4 . **Answer:** 5.2×10^{-4} ; 6×10^7

EXTRA: A cell phone has 6.4×10^{10} bytes of storage. Write it in standard form. **Answer:** 64,000,000,000

CULTURE CONNECTION — MINNESOTA STORIES

Spotlight: Ethiopian (Ethiopia)

Ethiopian Americans are a growing Minnesota community, centered in Cedar-Riverside. Ethiopian academic tradition dates to antiquity, including the Ge'ez numeral system used for counting and record-keeping.

Math Word Problem: An Ethiopian scribe records the number of star-grain seeds counted in Ge'ez as 4×10^8 . Write 4×10^8 in standard form, and state how many times larger it is than 4×10^3 .

CAREER SPOTLIGHT — MINNESOTA

Career: Best Buy — Richfield | Inventory Specialist | HS | median \$37,330 MN DEED 25th-percentile wage; OES-cited. Connection: algebra skills used daily on the job.

DIFFERENTIATION

- Scaffolding: sentence starters for translation tasks; GEMDAS reference card on desks.
- Extension: students write and solve their own word problems using the lesson's operation keywords.
- Language support: bilingual vocabulary cards; pair students with a math-language peer.
- Grouping: flexible pairs/trios by readiness; Merit Time games rotate roles each round.

ASSESSMENT

- Formative: whiteboard checks during the bell ringer; circulation during GO/SET/EXTRA; exit ticket each lesson.
- Game-based: Merit Time games give instant fluency feedback (L1–L5).
- Summative: Quiz 1 (Day 7, L1–L5), Quiz 2 (Day 17, L6–L15), End-of-Unit Test (Day 20, L1–L17).